



FB MSBS Grot

The **FB MSBS Grot** (Polish: *Modułowy System Broni Strzeleckiej „Grot”*, English: Modular Firearm System "Spearhead") is a family of modular assault rifle developed and manufactured by FB "Łucznik" Radom. Early prototypes were known as the *MSBS-5.56 Radon*.

There are two basic variants of the MSBS Grot: a conventional configuration assault rifle and a bullpup configuration.^[2] The MSBS is the first machine carbine fully designed and produced completely in Poland since World War II as well as the largest firearms sale contract taken on by the Polish arms industry since 1989.^[3]

History

The MSBS program was in development since 2007 by the Military University of Technology (WAT) in Warsaw with cooperation from the firearms manufacturer FB Radom,^{[4][5]} and was designed to replace the FB Beryl rifle used by the Polish military since 1997.

During initial tests conducted by the Military University of Technology, the MSBS-5.56 and its counterpart HK416 were subjected to comparative testing of the barrel and hand-guard for overheating during an intensive firing schedule. According to the tests, MSBS yielded the better results of the two rifles despite having a relatively light and longer barrel profile.^[6] Additional tests have also shown that MSBS is less susceptible to jamming when using lower quality ammunition or when it is insufficiently maintained,^[7] and had a more manageable recoil when fired in full-auto

FB MSBS Grot	
	
MSBS Grot C16 FB-M1	
Type	<u>Assault rifle</u> (MSBS Grot C) <u>Bullpup assault rifle</u> (MSBS Grot B)
Place of origin	<u>Poland</u>
Service history	
Used by	See <i>Users</i>
Wars	<u>Russian invasion of Ukraine</u> ^[1]
Production history	
Designer	Adam Gawron, Bartosz Stefaniak, Grzegorz Misiólek, Maciej Sajdak
Designed	2007
Manufacturer	<u>FB "Łucznik" Radom</u>
Produced	2007–2017 (prototypes) 2018–present
Variants	See <i>Variants</i>
Specifications	
Mass	3.7 kg (8.2 lb) (Grot C) 3.4 kg (7.5 lb) (Grot B)
Length	980 mm (39 in) (Grot C, stock extended) 720 mm (28 in) (Grot B)
Barrel length	254 mm (10.0 in) 406 mm (16.0 in) 508 mm (20.0 in) 267 mm (10.5 in) (Grot S) 368 mm (14.5 in) (Grot S)
Width	65 mm (2.6 in)
Cartridge	<u>5.56×45mm NATO</u> <u>7.62×39mm</u>

compared to other assault rifles chambered in the 5.56 cartridge, such as FB Beryl, HK416, FN SCAR-L or CZ BREN 2.^[8]

One of the main design characteristics of the MSBS-5.56 is the modular weapon system. If necessary, a soldier can sacrifice, for example, one standard assault rifle to quickly repair a light machine gun during a firefight. Another feature of the MSBS Grot is barrel changeability, as found on the Steyr AUG. This means a soldier may be able to adapt their rifle based on the environment they are in, such as converting a carbine to a designated marksman rifle, in order to engage targets at a greater range. The benefits of this feature allow an infantry squad to have fewer specialised weapons.

MSBS may also be used with components other than the ones manufactured domestically, and it is the first Polish rifle fully compliant with the NATO standards. The MSBS-5.56 features fully ambidextrous controls and Picatinny rail, and it may also be fed from standard STANAG magazines.^[9]

Currently, the rifle is in service with the Polish Territorial Defence Forces and is scheduled to replace FB Beryl as the primary infantry weapon and service rifle of the Polish Armed Forces.^[10] Following the adoption of the MSBS-5.56 by the Polish military, the *Grot* moniker has been added to the MSBS name in honor of the Home Army commander, General Stefan "Grot" Rowecki.^{[10][11]}

During the 2022 Russian invasion of Ukraine 10,000 Grot C16A2 rifles were sent to the Ukrainian military.^{[1][12][13]}

On May 9, 2022, the public was informed about the conclusion of an agreement between the Polish Armaments Group and an undefined East African country for the supply of an unspecified number of MSBS Grot assault rifles chambered for 7.62×39mm rounds as specified by the customer, along with 40 mm grenade launchers and optical sights DCM-1 Szafir^[14] manufactured by PCO S.A.^[15]

Variants

The following variants were designed based on the requests from the Polish Armed Forces to replace currently used firearms such as the AKM, FB Beryl, FB Mini-Beryl, and Pallad grenade launcher.

The MSBS GROT is a selective fire modular assault rifle which is capable of semi-automatic, 3-round burst and fully automatic fire. It has a cyclic rate of fire of around 700–900 rounds per minute.

Currently only the MSBS Grot C in assault rifle, carbine, configuration with a grenade launcher, MSBS Grot R and MSBS Grot S are available. The MSBS Grot B, and the machine gun and designated marksman rifle configurations are not yet produced. Conversion kits between variants are also not yet available. Conversion kit to change the calibre to 7.62×39mm is available.^[16]

Action	<u>Short-stroke gas piston</u> , <u>rotating bolt</u>
Rate of fire	700–900 rounds/min
Muzzle velocity	870 m/s (2,900 ft/s)
Effective firing range	500–800 m (547–875 yd)
Feed system	30-round detachable 5.56×45mm <u>STANAG</u> NATO magazines, 60-round <u>casket</u> <u>magazine</u> 30-round 7.62×39mm <u>STANAG</u> magazine
Sights	Integrated <u>Picatinny rail</u> for various <u>optical sights</u> and <u>Picatinny attachable iron</u> <u>sights</u>

- **MSBS Grot C** (Polish: *klasyczny*, English: classic) is the classic configuration of the MSBS-5.56, chambered in either 5.56×45mm NATO or 7.62×39mm cartridge with a folding and retractable stock.^[17]
 - MSBS-5,56A0 - initial production, 1,000 produced.
 - **FB-M1** (MSBS-5,56A1) - first production batch: modified charging handle and additional strap mount point on the picatinny rail.^[18]
 - **FB-M2** (MSBS-5,56A2) - second production batch: longer handguard to cover the gas block, reinforced firing pin for dry fire practice, improved pistol grip and buttstock.^[19]
- **MSBS Grot B** (Polish: *bezkolbowy*, English: bullpup) – is similar to the MSBS Grot C but in bullpup configuration. The MSBS Grot was designed to share a universal upper receiver for both the classic and bullpup configurations, which leads to the ease of conversion by simply swapping the lower receiver.
- **MSBS Grot R** – is the representative variant of the MSBS Grot designed to be used by honor guards which has been adapted to fire blanks and withstand drills. With the modularity of the MSBS Grot, the barrel can be swapped to fire live ammunition.
- **MSBS Grot S** – civilian semi-automatic only variant of the MSBS Grot without the bayonet-mount, available with 4 different variants of barrels: 10.5-, 14.5-, and 16-inch in .223 Remington and 16-inch in 7.62×39mm.^[20]



Prototypes of the MSBS-5.56 during the development stage in 2011 in both the conventional and bullpup layout



MSBS Grot B and MSBS Grot C



Soldiers of the Territorial Defence Force (WOT) with MSBS Grot C16



WOT soldiers with MSBS Grot C16 and Javelin missile launcher

Both the MSBS Grot C and MSBS Grot B are planned in 5 different configurations and have many common interchangeable parts and can be converted from one to the other:^{[21][22]}

- Assault rifle – a standard assault rifle configuration with 16 in (410 mm) barrel.
- Assault rifle with underbarrel grenade launcher – similar to the assault rifle configuration but with a 40×46mm underbarrel grenade launcher equipped.
- Carbine – a short barrel variant with a 10 in (250 mm) barrel.
- Carbine with underbarrel grenade launcher – similar to the carbine configuration but with a 40×46mm underbarrel grenade launcher equipped.
- Machine gun – essentially an automatic rifle variant for both Grot C and Grot B. It features a heavier barrel fitted with a flash hider, and the bayonet-mount was removed. This variant is meant to be used with high-capacity magazines. Surefire MAG5-60 were used for demonstration purposes.^[23]
- Designated marksman rifle – a designated marksman rifle variant fitted with a lighter and more precise 16 in (410 mm) and 20 in (508 mm) barrel, and has a two stage trigger group.

Polish military adoption

Since the Polish ministry of defence ordered 53,000 of the MSBS assault rifles, Fabryka Broni has introduced a military naming designation for the MSBS series.^[24]

Grot – After the official adoption of the MSBS-5.56 by the Polish Armed Forces, the Grot name was added.

C, B, R, S – *variants*: classic, bullpup, representative, sport (semi-auto civilian). Denoted after the Grot designation.

10, 16, 20 – *barrel lengths*, in inches. So far only the 16 in (406 mm) has been adopted by the Polish military.

C, G, M, PS – *configurations*: carbine, variants with grenade launcher, automatic rifle, designated marksman rifle. Denoted after the barrel length designation.

FB – Fabryka Broni, the manufacturer of this weapon system.

M(x) – number of series (x stands for the generation, for examples, *M1* means first generation, *M2* is second generation, *M3* is planned third generation).

The *M2* was developed to address the issues found on the early production models.^{[25][26]} The bullpup configuration did not receive the *M2* upgrade due to the lack of interest from the Polish military. The changes included a longer upper receiver with M-LOK compatible handguard and heat stop covers, a quick-detach sling mounting points, FB made pistol grip instead of previous Magpul pistol grip due to shortage during mass production, reinforced buttstock, reinforced secondary sights, strengthened, thicker charging handle, reinforced firing pin (not interchangeable with the old one from *M1*), FB made polymer translucent magazine, new lock (not interchangeable with the old one from *M1*), and new slide (not interchangeable with the old one from *M1*).^[27]

Grot 762N

Grot 762N is a semi-automatic designated marksman rifle chambered in 7.62×51mm NATO cartridge.

The MSBS-7.62N project began in the end of 2015 when Polish Military of Defence announced analytical works for the replacement program of their SVD Dragunov marksman rifle and supplement the bolt-action TRG-22 and Tor sniper rifles currently in service. It was led by the team of Fabryka Broni Łucznik-Radom and Wojskowa Akademia Techniczna (WAT) engineers.^[28]

The Grot 7.62N has two different configurations; one with a 508 mm (20 in) barrel with a fixed adjustable stock and the

Grot 762N	
Type	<u>Designated marksman rifle</u>
Place of origin	Poland
Production history	
Manufacturer	FB "Łucznik" Radom
Specifications	
Mass	5.2–6 kg (11–13 lb)
Barrel length	406 mm (16.0 in) 508 mm (20.0 in)
Cartridge	<u>7.62×51mm NATO</u>
Action	Short-stroke gas piston, rotating bolt
Feed system	20-round detachable <u>SR-25</u> magazines

other with 406 mm (16 in) barrel with the MSBS Grot C's adjustable side-folding stock. Both configurations are part of the MSBS family and some of the parts of the Grot 762N, such as the stock, handguard, pistol grip, and trigger are interchangeable between 5.56mm and 7.62mm models.^{[28][29]}

Sights	Integrated <u>Picatinny rail</u> for various optical sights and Picatinny attachable <u>iron sights</u>
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The Grot 7.62N is designed as a semi-automatic rifle but project engineers declared that a fully automatic configuration can be developed if required.^[28]

Users

-  Poland: Polish Armed Forces, 184,000 planned by 2026.^{[30][31]}
 - Territorial Defence Force, 53,000 in 2017.^[30]
-  Rwanda: On September 1, 2023, PGZ concluded a contract for the supply of MSBS Grot rifles with an unspecified East-Central African country.^[32] On January 31, 2024, the Defense24 portal listed Rwanda as the recipient of Polish-made rifles, but it is not certain whether the deliveries concern Grot or other rifles.^[33] In 2024, photos circulated on the web showing a Rwandan soldier with a 7,62x39 Grot fitted with a GP grenade launcher ^[34]
-  Ukraine: 10,000 Grot C16 FB-M1 and A2 variant assault rifles sent to the Ukrainian Ground Forces during the 2022 Russian invasion of Ukraine.^[1] In 2023, it was announced that an unspecified number of Grot C16 A2 rifles had been ordered by Ukraine, but later on such claims have been denied.^{[35][36]}
-  United States: unknown quantity sold to Special Forces Command (USSOCOM).^[37]

See also

- Remington ACR
- FB Beryl

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External links

- Fabryka Broni "Łucznik" Radom home page (<http://www.fabrykabroni.pl/>) Archived (<https://web.archive.org/web/20070312171410/http://www.fabrykabroni.pl/>) 12 March 2007 at the Wayback Machine
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